- See blog post on grades as a prospective professor.

- Use whatever resources I want for my learning. Stack Exchange is good. Cite whatever resources I used. I don't need to if I talk to Gabby (the TA) or the professor.

- Avoid sterio types on who is and is not good at math. Be kind when you're correcting someone and work on being vulnerable. Treat them with respect.

- Scan and upload my notes to the professor after each class.

- Abstract Algebra 1/22/26

Administrative annoucements

- HW due on Thursdays! - The In-Person Colloquium (SMIMIC) is coming up and is jam packed. Feb. 5th - April 23rd, there will be talks on math topics
- Free pizza
- Prof. Smith is co-chairing this commitee, so he's extra busy and may be slow to respond to emails

- HW Questions

Notation: $\mathbb{Z}/m\mathbb{Z}$ "z mod m z" $\mathbb{Z}$ mod $m\mathbb{Z}$

$\mathbb{Z}/m\mathbb{Z} = \mathbb{Z}/m = \mathbb{Z}/\langle m \rangle$

(7) Modulo 2: $1^2+1 = 2 \equiv 0 \bmod 2$ 1/22/26

$3^2+1 = 10 \equiv 0 \bmod 2$

$(2n+1)^2+1 = 4n^2+4n+2 \equiv 0 \bmod 2$, for some $n \in \mathbb{Z}$.

We partion the integers into equivalence classes/ congruence classes/ residue classes

$[0]_2 = \{ n \in \mathbb{Z} : n \equiv 0 \bmod 2 \}$.

$= \{ n \in \mathbb{Z} : \underbrace{2 \mid n}_{\text{divides}} \} = [2]_2 = [4]_2$
$= [-278418]_2$
$= \bar{0} = \bar{2} = \bar{4}$

$\equiv$ vs. $=$

congruance — equals

the set is always the integers — used when talking about equivalent classes in $\mathbb{Z}/m\mathbb{Z}$.

E.g. $16 \equiv -1 \bmod 17$ but $16 \neq -1$.

$[16]_{17} = [-1]_{17} = \{ n \in \mathbb{Z} : n \equiv -1 \bmod 17 \}$

$\parallel$ $\qquad$ $\parallel$ ✓

$\{ n \in \mathbb{Z} : n \equiv 16 \bmod 17 \}$

$$[x]_m^2 + [1]_m = [0]_m$$
$$\Rightarrow [x^2]_m + [1]_m = [x^2+1]_m = [0]_m$$

Lesson: Chap. 2

Groups

this implies closure

Let G be a set and let $* : G \times G \to G$ be a well-defined binary opperation.

We say G is a group if the following are satisfied:

1) For any $a, b, c \in G$, $(a * b) * c = a * (b * c)$, called associativity.

2) $\exists \underbrace{e}_{\text{id. or } 1} \in G$ such that $e * a = a * e = a$, $\forall a \in G$.
- the identity element exists for $*$

3) For every $a \in G$, there exists an inverse element in G, $a^{-1} \in G$, such that $a * a^{-1} = a^{-1} * a = e$
- every element in G has an inverse element.

If in addition G is commutative under $*$, then we say that G is Abelian.
i.e., $\forall a, b \in G$, $a * b = b * a$.

Subtraction is not commutative and associativity also fail (often).

Matrix multiplication is not commutative 1/22/26

Examples

$\underbrace{\mathbb{Z} \text{ under addition}}$ is an Abelian group

$(\mathbb{Z}, +)$

$(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$ } All Abelian groups

$(\mathbb{Z}/2\mathbb{Z}, +)$, $(\mathbb{Z}/m\mathbb{Z}, +)$,

(2x2 matricies in $\mathbb{R}$ under addition)

$(\mathbb{Z}, \cdot)$ is not a group b/c (for example) 2 has no inverse).

$(\mathbb{Q}, \cdot)$ is not a group b/c 0 has no mult. inverse element.

$(\mathbb{Q} - \{0\}, \cdot)$ <u>is</u> an Abelian group.

$(\underbrace{\mathbb{Q}^+}, \cdot)$ is too.

the possitive rationals

$\mathbb{Q} - \{0\} = \mathbb{Q}^\times = \mathbb{Q}^* = \mathcal{U}(\mathbb{Q})$

= the elements of $\mathbb{Q}$ w/ mult. inverses.

$(\mathbb{R}^\times = \mathbb{R} - \{0\}, \cdot)$ is an Abelian Group.

$$\mathbb{Z}^{\times} = \{ n \in \mathbb{Z} : \exists n^{-1} \in \mathbb{Z} \text{ s.t. } n \cdot n^{-1} = 1 \}$$

$$= \{ 1, -1 \} = \{ \pm 1 \}.$$

We can prove directly that $(\{\pm 1\}, \cdot)$ is Abelian group.

$\cdot$	1	-1
1	1	-1
-1	-1	1

Kally Table